

LINEAR Support Vector Machines

Linear SVM

- A Linear Support Vector Machine is a supervised machine learning algorithm used for binary classification. It finds the optimal separating hyperplane (a straight line in 2D, a flat plane in 3D, and so on) that divides two classes while maximizing the margin — the empty buffer zone between the boundary and the nearest data points of each class.
- *Linear SVM draws the single straight line that is as far away as possible from the nearest points of both classes, and those nearest points are the support vectors that pin the line in place.*

The Linear SVM Formula

Just like the Perceptron, the Linear SVM makes decisions using a weighted sum of the inputs plus a bias, called the decision function:

- $f(x) = w \cdot x + b = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$

The predicted class is then found using the sign function instead of a step function:

- $y = \text{sign}(w \cdot x + b)$
- **+1** if $w \cdot x + b \geq 0$
- **-1** if $w \cdot x + b < 0$

Breakdown of the Formula

Symbol	Name
$x = (x_1, \dots, x_n)$	Feature vector
$w = (w_1, \dots, w_n)$	Weight vector
b	Bias / Intercept
$w \cdot x + b$	Decision function
$\text{sign}()$	Sign function
y_i	True class label
$\ w\ $	Norm (magnitude) of w
Support vectors	—

The Margin

SVM does not just draw one line — it conceptually draws three parallel lines: the decision boundary in the middle, and two margin boundaries on either side, touching the nearest points of each class:

- **Positive margin boundary: $w \cdot x + b = +1$**
- **Negative margin boundary: $w \cdot x + b = -1$**

The distance between these two margin boundaries is the margin, and it is calculated directly from the length of w :

- **Margin width = $2 / \|w\|$**

How Linear SVM Separates the Classes

The Decision Boundary and the Margin “Street”

- The equation $w \cdot x + b = 0$ is the decision boundary, exactly as in the Perceptron. But SVM adds two more parallel lines around it ($w \cdot x + b = 1$ and $w \cdot x + b = -1$) that mark the edges of the widest possible empty street between the classes.

Support Vectors: The Points That Actually Matter

- Once a Linear SVM is trained, most of the training points turn out to be irrelevant to the final boundary — only the points that sit exactly on the margin boundaries (where $y_i(w \cdot x_i + b) = 1$) are the support vectors.

Separating the Class Fields in Your Dataset

Example 1: Classifying Points with a Trained Linear SVM

Point	x1	x2	y (target class)
P1	0	1	-1
P2	1	3	-1
P3	4	2	+1
P4	5	4	+1

Given trained weights: $w = (1, 1)$, bias $b = -5$

Step 1: Write the decision function for this problem:

$$f(x) = (1)(x1) + (1)(x2) + (-5)$$

$$f(x) = x1 + x2 - 5$$

Step 2: Substitute each point's feature values and apply the sign function:

Example 1: Classifying Points with a Trained Linear SVM

Point	Computation: $f(x) = x_1 + x_2 - 5$	$f(x)$	Predicted $y = \text{sign}(f(x))$	Matches target?
P1	$0 + 1 - 5$	-4	-1	Yes
P2	$1 + 3 - 5$	-1	-1	Yes
P3	$4 + 2 - 5$	1	+1	Yes
P4	$5 + 4 - 5$	4	+1	Yes

Step 3: Separate the classes.

The decision boundary is $x_1 + x_2 - 5 = 0$, or $x_2 = 5 - x_1$.

The **sign of $f(x)$** determines the label:

- $f(x) < 0 \rightarrow -1$
- $f(x) > 0 \rightarrow +1$

Example 2: Identifying the Support Vectors

Point	y (label)	f(x)	Functional margin: $y \times f(x)$	Support vector?
P1	-1	-4	$(-1)(-4) = 4$	No — far inside safe zone
P2	-1	-1	$(-1)(-1) = 1$	Yes — sits exactly on the margin
P3	+1	1	$(+1)(1) = 1$	Yes — sits exactly on the margin
P4	+1	4	$(+1)(4) = 4$	No — far inside safe zone

Step 1: For every point, compute the functional margin:

$$y_i \times f(x_i) = y_i \times (x_1 + x_2 - 5)$$

A value of exactly 1 means the point sits precisely on its margin boundary.

Step 2: Interpret the result.

Example 3: Computing the Margin Width

Point	x1	x2	y (target class)
P1	0	1	-1
P2	1	3	-1
P3	4	2	+1
P4	5	4	+1

Given trained weights: $w = (1, 1)$, bias $b = -5$

Step 1: Compute the norm (length) of the weight vector:

$$\|w\| = \sqrt{w_1^2 + w_2^2} = \sqrt{1^2 + 1^2}$$

$$= \sqrt{2} \approx 1.414$$

Step 2 : Apply the margin formula:

$$\text{Margin width} = 2 / \|w\|$$

$$= 2 / 1.414 \approx 1.414$$

Example 3: Computing the Margin Width

Point	x1	x2	y (target class)
P1	0	1	-1
P2	1	3	-1
P3	4	2	+1
P4	5	4	+1

Given trained weights: $w = (1, 1)$, bias $b = -5$

Step 3: Write the two margin boundary equations using $b = -5$:

Positive margin boundary:

- $x_1 + x_2 - 5 = 1$
- $x_1 + x_2 = 6$

Negative margin boundary:

- $x_1 + x_2 - 5 = -1$
- $x_1 + x_2 = 4$

Step 4: Interpret.